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SEALING A PORTION OF A WELLBORE

Abstract

A cement swell packer includes a base tubular that includes at least one end connection; a cement swell assembly configured to mount on the base tubular and including a portion of a dry cementitious material configured to form a hardenable or curable cement when exposed to a fluid in a wellbore, and a mesh configured to hold the portion of the dry cementitious material; and a liquid barrier formed over the dry cementitious material.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION [0001] This application claims priority under 35 U.S.C. § 119 to U.S. Provisional Patent Application Ser. No. 63/560,969, filed on Mar. 4, 2024, the entire contents of which are incorporated by reference herein.

TECHNICAL FIELD

[0002] The present disclosure describes apparatus, systems, and methods associated with sealing a portion of a wellbore and, more particularly, sealing a portion of a wellbore with a cement swell packer.

BACKGROUND

[0003] Conventional wellbore cementing is an important process in well construction that involves the placement of cement into the annular space between the casing and the wellbore wall. The process serves multiple purposes, including providing zonal isolation, ensuring well integrity, and facilitating efficient production or injection of fluids. During cementing, a specially formulated cement slurry is pumped down the casing and forced up into the annular space. The cement slurry then sets and hardens, creating a durable and impermeable barrier that prevents the migration of fluids between different zones and protects the wellbore from potential environmental or operational risks. Cementing plays a vital role in achieving safe and efficient well operations, enhancing reservoir performance, and maintaining the integrity of all wells.

SUMMARY

[0004] In an example implementation, a cement swell packer includes a base tubular that includes at least one end connection; a cement swell assembly configured to mount on the base tubular and including a portion of a dry cementitious material configured to form a hardenable or curable cement when exposed to a fluid in a wellbore, and a mesh configured to hold the portion of the dry cementitious material; and a liquid barrier formed over the dry cementitious material.

[0005] An aspect combinable with the example implementation further includes at least one collar configured to retain the cement swell assembly on the base tubular.

[0006] In another aspect combinable with one, some, or all of the previous aspects, the at least one collar includes a first collar configured to mount on the base tubular at a first end of the cement swell assembly and a second collar configured to mount on the base tubular at a second end of the cement swell assembly opposite the first end.

[0007] In another aspect combinable with one, some, or all of the previous aspects, the liquid barrier includes a removable film or wrap.

[0008] In another aspect combinable with one, some, or all of the previous aspects, the liquid barrier includes a sprayed barrier configured to degrade in the presence of the fluid.

[0009] Another aspect combinable with one, some, or all of the previous aspects further includes at least one centralizer configured to mount over the base tubular.

[0010] In another aspect combinable with one, some, or all of the previous aspects, the at least one centralizer includes a first centralizer configured to mount on the base tubular at or near a first end of the base tubular and a second centralizer configured to mount on the base tubular at or near a second end of the base tubular opposite the first end.

[0011] In another aspect combinable with one, some, or all of the previous aspects, the base tubular includes a casing joint of a production casing.

[0012] In another aspect combinable with one, some, or all of the previous aspects, the at least one end connection includes a male threaded connection at a first end of the casing joint; and a female threaded connection at a second end of the casing joint opposite the first end.

[0013] In another aspect combinable with one, some, or all of the previous aspects, an unconfined compressive strength of the hardenable or curable cement is less than an unconfined compressive strength of a primary cement pumped into the wellbore to adhere the production casing to a well surface.

[0014] In another aspect combinable with one, some, or all of the previous aspects, the well surface

includes an intermediate casing.

[0015] In another example implementation, a method includes positioning a cement swell packer adjacent a wellbore entry. The cement swell packer includes a base tubular that includes at least one end connection, a cement swell assembly mounted on the base tubular and including a portion of a dry cementitious material configured to form a hardenable or curable cement when exposed to a fluid in a wellbore; and a mesh configured to hold the portion of the dry cementitious material, and a liquid barrier formed over the dry cementitious material. The method includes removing the liquid barrier from the dry cementitious material; making up a connection between the at least one end connection and another base tubular within a string of base tubulars; and running the cement swell packer into the wellbore on the string of base tubulars.

[0016] An aspect combinable with the example implementation further includes making up another connection between the base tubular and a further base tubular at an end of the base tubular opposite the another base tubular; and running the base tubular, the another base tubular, and the further base tubular into the wellbore on the string of base tubulars.

[0017] In another aspect combinable with one, some, or all of the previous aspects, removing the liquid barrier from the dry cementitious material occurs subsequent to making up the connection between the at least one end connection and the another base tubular within the string of base tubulars.

[0018] Another aspect combinable with one, some, or all of the previous aspects further includes positioning the base tubular on the string of base tubulars at a particular location in the wellbore; and exposing the dry cementitious material to a liquid in the wellbore to form a hardenable cementitious slurry.

[0019] Another aspect combinable with one, some, or all of the previous aspects further includes initiating a hardening or curing process of the hardenable cementitious slurry to form hardened cement over a time duration.

[0020] In another aspect combinable with one, some, or all of the previous aspects, exposing the dry cementitious material to the liquid in the wellbore includes circulating the liquid from a terranean surface to the particular location in the wellbore.

[0021] In another aspect combinable with one, some, or all of the previous aspects, the liquid includes water.

[0022] Another aspect combinable with one, some, or all of the previous aspects further includes pumping a flow of a primary cement that includes the liquid downhole through the string of base tubulars and uphole through an annulus between the string of base tubulars and a well surface; and exposing the dry cementitious material to the liquid in the wellbore to form the hardenable cementitious slurry.

[0023] In another aspect combinable with one, some, or all of the previous aspects, the well surface includes an inner radial surface of an intermediate casing string, and the string of base tubulars includes a production casing.

[0024] Another aspect combinable with one, some, or all of the previous aspects further includes assembling the cement swell packer by mounting the cement swell assembly on the base tubular; installing at least one collar onto the base tubular to retain the cement swell assembly on the base tubular; and forming the liquid barrier onto the cement swell assembly.

[0025] In another aspect combinable with one, some, or all of the previous aspects, forming the liquid barrier onto the cement swell assembly includes covering the cement swell assembly with a film or wrapping to form the liquid barrier; or spraying the cement swell assembly to form the liquid barrier.

[0026] In another example implementation, an apparatus includes a plurality of casing joints. Each casing joint includes a tubular section, a male threaded connection, and a female threaded connection. The plurality of casing joints are configured to form a connected production casing string in a wellbore. The apparatus includes a cement swell assembly mounted on at least one of the

casing joints. The cement swell assembly includes a portion of a dry cementitious material configured to form a hardenable or curable cement when exposed to a fluid in the wellbore; and a mesh configured to hold the portion of the dry cementitious material.

[0027] An aspect combinable with the example implementation further includes a first collar mounted on the at least one casing joint at a first end to retain the cement swell assembly on the casing joint; and a second collar mounted on the at least one casing joint at a second end of the cement swell assembly opposite the first end to retain the cement swell assembly on the casing joint.

[0028] Another aspect combinable with one, some, or all of the previous aspects further includes a first centralizer mounted on the casing joint at or near the first end; and a second centralizer mounted on the casing joint at or near the second end.

[0029] Another aspect combinable with one, some, or all of the previous aspects further includes a liquid barrier that covers the portion of the dry cementitious material.

[0030] In another aspect combinable with one, some, or all of the previous aspects, an unconfined compressive strength of the hardenable or curable cement is less than an unconfined compressive strength of a primary cement pumped into the wellbore to install the connected production casing string.

[0031] The details of one or more implementations of the subject matter described in this disclosure are set forth in the accompanying drawings and the description below. Other features, aspects, and advantages of the subject matter will become apparent from the description, the drawings, and the claims.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0032] FIG. 1 is a schematic diagram of an example implementation of a wellbore system that includes at least one cement swell packer installed in a tubular string according to the present disclosure.

[0033] FIG. 2 is an exploded schematic diagram of an example implementation of a cement swell packer according to the present disclosure.

[0034] FIG. 3 is a schematic diagram of an example implementation of an assembled cement swell packer according to the present disclosure.

[0035] FIG. 4 is a schematic diagram of an example implementation of a cement swell packer readied for installation into a wellbore according to the present disclosure.

[0036] FIG. 5 is a graph that shows relative strength development between a cement swell packer and a primary cement according to the present disclosure.

[0037] FIGS. 6A-6C show schematic diagrams that illustrate a development over time of a cement swell packer in a wellbore according to the present disclosure.

[0038] FIGS. 7A-7C show additional schematic diagrams that illustrate a development over time of a cement swell packer in a wellbore according to the present disclosure.

[0039] FIGS. 8A-8D show additional schematic diagrams that illustrate a development over time of a cement swell packer in a wellbore according to the present disclosure.

DETAILED DESCRIPTION

[0040] The present disclosure describes example implementations of a cement swell packer that can be installed within a wellbore tubular string, such as a casing string (e.g., production casing string or other casing string), to create a cementitious seal between the string and, for instance, the subterranean formation through which the wellbore is formed. In some aspects, example implementations according to the present disclosure can provide for an enhanced seal between, for example, the production casing and the subterranean formation in locations within the wellbore

where a primary cement (circulated to install the casing to the formation) is inadequate or non-existent.

[0041] FIG. 1 is a schematic diagram of an example implementation of a wellbore system **10** that includes at least one cement swell packer **100** installed in a tubular string according to the present disclosure. Generally, FIG. 1 illustrates at least a portion of one implementation of the cement swell packer **100** according to the present disclosure in which one or more cement swell packers **100** can be installed in a wellbore as part of a tubular string, such as, for example, a production casing **45** within a wellbore **20**. However, cement swell packers **100** according to the present disclosure can also be installed on other wellbore tubular strings, such as other casings, production tubing strings, or otherwise.

[0042] Example implementations of the cement swell packer **100** according to the present disclosure can provide for solutions to problems usually associated with inadequate or non-existent cementing of a casing. For example, cement swell packers according to the present disclosure can improve zonal isolation. Achieving effective zonal isolation can be crucial to prevent fluid migration between different formations and to protect a wellbore. Challenges arise from the need to properly design and execute a cementing process to ensure a complete and uniform cement sheath, especially in challenging downhole conditions.

[0043] Also, cement swell packers **100** according to the present disclosure can improve cement slurry stability. Maintaining the stability and consistency of a primary cement slurry throughout a pumping operation can be essential. Factors such as temperature, pressure, and wellbore conditions can impact the slurry's properties, leading to issues like premature setting, fluid loss, or poor bonding with the casing or formation (or both).

[0044] Further, cement swell packers **100** according to the present disclosure can improve cement placement. Proper placement of a cement slurry is critical for achieving effective zonal isolation. Challenges can arise from complex well geometries, narrow annular spaces, or formations with varying permeability, which can result in uneven distribution of the cement, voids, or channels that compromise zonal isolation.

[0045] As another example, cement swell packers **100** according to the present disclosure can improve wellbore integrity. Cement integrity and bonding between the casing and formation can be essential for maintaining wellbore integrity. Challenges may arise from factors such as inadequate cement coverage, poor bonding, or the presence of micro-annuli, which can lead to fluid migration, gas influx, or mechanical failures, such as casing collapse or casing leaks.

[0046] Further, cement swell packers **100** according to the present disclosure can help avoid problems with wellbore fluids and additives. For example, the presence of certain wellbore fluids, such as drilling mud or high-salinity brines, can pose challenges during cementing. Incompatibilities between these fluids and the primary cement slurry can result in issues like fluid loss, reduced cement performance, or compromised zonal isolation. Additionally, selecting appropriate additives and their proper dosage can be crucial to address specific wellbore conditions and challenges effectively.

[0047] In this example implementation, the cement swell packer **100** (or many cement swell packers **100**) are installed as part of the production casing **45** into the wellbore **20**, which is used to circulate a wellbore fluid **65**, such as a hydrocarbon fluid (for example, oil, gas, or a mixture thereof) from a subterranean reservoir **40**, through one or more perforations **55** formed in the production casing **45** (as well as fractures, whether hydraulic or natural or both), and to a terranean surface **12**. As shown, the wellbore **20** accesses the subterranean formation **40** and provides access to hydrocarbons (for example, the wellbore fluid **65**) located in such subterranean formation **40**. In an example implementation of system **10**, the system **10** may be used for a production operation in which the hydrocarbons may be produced from the subterranean formation **40** through the production casing **45** (and also, in some cases, other wellbore tubular strings) and to the surface **12**.

[0048] A drilling assembly (not shown) may be used to form the wellbore **20** extending from the

terranean surface **12** and through one or more geological formations in the Earth. One or more subterranean formations, such as subterranean zone **40**, are located under the terranean surface **12**. As will be explained in more detail below, one or more wellbore casings, such as a surface casing **30**, intermediate casing **35**, and the production casing **45** can be installed in at least a portion of the wellbore **20**. In some aspects, production casing **45** is nested within the intermediate casing **35** such that an annulus **22** is formed between the production casing **45** and intermediate casing **35**; however, the annulus can also be defined between the production casing **45** and wellbore **20** (i.e., the rock formation) when these two casings (**35** and **45**) are not nested.

[0049] In some implementations, a drilling assembly used to form the wellbore **20** may be deployed on a body of water rather than the terranean surface **12**. For instance, in some implementations, the terranean surface **12** may be an ocean, gulf, sea, or any other body of water under which hydrocarbon-bearing formations may be found. In short, reference to the terranean surface **12** includes both land and water surfaces and contemplates forming and developing one or more downhole pumping systems **10** from either or both locations.

[0050] In some implementations of the wellbore system **10**, the wellbore **20** is cased with one or more casings. As illustrated, the wellbore **20** includes a conductor casing **25**, which extends from the terranean surface **12** shortly into the Earth. A portion of the wellbore **20** enclosed by the conductor casing **25** may be a large diameter borehole. Additionally, in some implementations, the wellbore **20** may be offset from vertical (for example, a slant wellbore). Even further, in some implementations, the wellbore **20** may be a stepped wellbore, such that a portion is drilled vertically downward and then curved to a substantially horizontal wellbore portion. Additional substantially vertical and horizontal wellbore portions may be added according to, for example, the type of terranean surface **12**, the depth of one or more target subterranean formations, the depth of one or more productive subterranean formations, or other criteria.

[0051] Downhole of the conductor casing **25** may be the surface casing **30**. The surface casing **30** may enclose a slightly smaller borehole and protect the wellbore **20** from intrusion of, for example, freshwater aquifers located near the terranean surface **12**. The wellbore **20** may then extend vertically downward. This portion of the wellbore **20** may be enclosed by the intermediate casing **35** or the production casing **45** (or both). Any of the illustrated casings, as well as other casings that may be present in the wellbore system **10**, may include one or more casing collars.

[0052] As shown in this example implementation, the production casing **45** is formed of multiple production casing joints **50** (also called casing joints **50**). Indeed, depending on a total length of the production casing **45**, there can be hundreds of casing joints **50** coupled together (for example, threadingly) to form the production casing **45**. Once the production casing **45** is installed in the wellbore, a primary cement **70** is circulated downhole through the installed production casing **45** and then back uphole in an annulus **22** between an outer radial surface of the production casing **45** and the wellbore **20** (i.e., the subterranean formation through which the wellbore **20** is formed). Generally, the primary cement **70** remains in the annulus **22** to adhere the production casing **45** to the formation(s) through which the wellbore **20** is formed. The primary cement **70** can also provide other functions, such as sealing the wellbore **20** (and annulus **22**) against fluid intrusion and providing better wellbore integrity (as compared to having no primary cement **70** between the production casing **45** and the formation(s)).

[0053] One or more cement swell packers **100** can be installed within the production casing **45** during a run-in process that places the production casing **45** (joint-by-joint) into the wellbore **20**. FIG. 2 is an exploded schematic diagram of an example implementation of a cement swell packer **100** according to the present disclosure. In this example implementation, the cement swell packer **100** includes a casing joint **50** onto which a cement swell assembly **101** is installed (prior to running into the wellbore **20** as part of the production casing **45**).

[0054] As shown in this example, the casing joint **50** is comprised of a tubular section **52** (e.g., of circular or near circular cross-section) that includes a female connection **54** and a male connection

56. In this example, both the female connection **54** and male connection **56** are threaded connections, with threads formed on an outer radial surface of the casing joint **50** for the male connection **56**, and threads formed on an inner radial surface of the female connection **54**. However, other forms of connections are possible, such as grooved joint connections (for example, Victaulic® or otherwise).

[0055] A diameter, d , of the female connection **54** generally represents the largest diameter in the casing joint **50** (i.e., relative to the diameter of the tubular section **52** and male connection **56**). The diameter, d , in this example, is generally set by conventional casing joint sizes of 13 $\frac{3}{8}$ " or 18 $\frac{5}{8}$ " (but other sizes are also contemplated by the present disclosure). In this example, the casing joint **50** can be 6 ft. or 10 ft. long, but any length is also contemplated by the present disclosure. Also, although shown as a solid tubular (and specifically, a tubular for casing), the casing joint **50** can represent a casing, a liner, a completion pipe, as well as tubular screens of the wire, mesh or pre-perforated variety.

[0056] In this example implementation of the cement swell packer **100**, the cement swell assembly **101** is comprised of a cementitious mesh packer **102** that is sized to ride on the tubular section **52** of the casing joint **50**. The cementitious mesh packer **102** includes a fabric or mesh **104** that is impregnated or embedded with, or otherwise holds or retains a dry cementitious material **106** that, when exposed to one or more fluids of sufficient volume, forms a hardenable or curable cement barrier. The dry cementitious material **106** can be, for example, a Portland cement granular material that, with the addition of liquid (such as water or brine in the wellbore or circulated into the wellbore) swells to form cement.

[0057] As further shown in this example, the cement swell assembly **101** includes an upper collar **108** and a lower collar **110**. Generally, the collars **108** and **110** ride on the tubular section **52** on either end of the cementitious mesh packer **102** to retain the cementitious mesh packer **102** on the tubular section **52** of the casing joint **50** (for example, during a run in of the cement swell packer **100** when installing the production casing **45** in the wellbore **20**).

[0058] In this example, a diameter, D , of the cementitious mesh packer **102** represents the largest diameter of the cement swell assembly **101** (which is generally, the diameter of the collars **108** and **110**) prior to swelling of the cementitious material **106** and therefore, the diameter of the cement swell assembly **101** when it is installed on the casing joint **50** and run into a wellbore. In some aspects, the diameter, D , can be equal to or less than the diameter, d , of the casing joint **50**. Thus, the cement swell assembly **101** installed on the casing joint **50** would not impede installation of the production casing **45** into the wellbore **20**. Given this dimension, D , being less than or equal to the dimension, d , the cement swell packer **100** is not expected to contact an inside diameter of the wellbore **20**. And with optional centralization (as described herein), even the upper collar **108** and lower collar **110** should not (or should rarely) contact the rock formation through which the wellbore **20** is formed. Further, it is not expected that the casing running speed will need to be altered to avoid wellbore challenges such as "surging."

[0059] In this example implementation of the cement swell assembly **101**, a barrier **112** can be installed over or around the cementitious mesh packer **102** during installation of the cement swell assembly **101** onto the casing joint **50**. Generally, the barrier **112** can be a mechanical barrier (for example, a film or wrapping such as a plastic or otherwise liquid impenetrable barrier) that protects the cementitious material **106** from, for example, liquids or other intrusive material or damage. The barrier **112** is removable (for example, prior to running the cement swell packer **100** into the wellbore **20**) to expose the cementitious mesh packer **102**. In alternative implementations of the barrier **112**, the barrier **112** can be a spray or other liquid barrier that is applied to the cementitious mesh packer **102** and remains on the cementitious mesh packer **102** during installation of the cement swell packer **100** into the wellbore **20**, but eventually wears off or dissolves in the presence of liquid to expose the cementitious material **106** to a liquid swelling fluid in the wellbore **20**.

[0060] FIG. **3** is a schematic diagram of an example implementation of an assembled cement swell

packer **100** according to the present disclosure. As shown in this example, the cement swell assembly **101** has been installed onto the casing joint **50** over the tubular section **52**. The cementitious mesh packer **102** rides on the tubular section **52** and is held in place by the upper collar **108** and lower collar **110**. Optional centralizers **120** can be installed on the casing joint **50** on either end of the cement swell assembly **101** as shown. In some aspects, one or more of the centralizers **120** can be added to the cement swell packer **100**, such as when there is concern that damage might occur to the cement swell assembly **101** during running into the wellbore **20** as part of the production casing **45**.

[0061] FIG. **3** can represent the cement swell packer **100** (with or without one or more centralizers **120**) upon delivery to a well site or at a well site but prior to making up the cement swell packer **100** into the production casing **45** and installing it into the wellbore **20**. For example, in this illustration, the barrier **112** is still installed on the cementitious mesh packer **102** and not yet removed to run the cement swell packer **100** into the wellbore **20**. In some aspects, the barrier **112** can be removed (if a mechanical barrier) just prior to making up the cement swell packer **100** into the production casing **45** on the rig floor. In some aspects, the barrier **112** can be removed (if a mechanical barrier) after making up the cement swell packer **100** into the production casing **45** on the rig floor but prior to running into the wellbore **20**.

[0062] FIG. **4** is a schematic diagram of an example implementation of the cement swell packer **100** readied for installation into a wellbore according to the present disclosure. For example, as shown in this figure, the cement swell packer **100** is coupled (for example, threadingly) to another casing joint **50** (which may or may not include a cement swell assembly **101**) and is being run into the wellbore **20**. In this example, the optional centralizers **120** are positioned on the tubular section **52** of the casing joint **50**. Further, in this example, the barrier **112** (as a mechanical barrier such as a film or wrap) has been removed from the cementitious mesh packer **102**. The barrier **112** can be removed prior to the cement swell packer **100** being made up with the casing joint **50** that is illustrated as being run into the wellbore **20**. Alternatively, the barrier **112** can be removed subsequent to the cement swell packer **100** being made up with the casing joint **50** that is illustrated as being run into the wellbore **20**.

[0063] In an example operation, once the cement swell packer **100** is assembled as described with reference to FIGS. **2** and **3** (either at a well site or off site), the cement swell packer **100** can be coupled within a production casing **45** that is being run into the wellbore **20** (as described with reference to FIG. **4**). If necessary, as described, the barrier **112** is removed prior to the cement swell packer **100** being run into the wellbore. In some aspects, the cement swell packer **100** (or multiple cement swell packers **100**) can be installed in a production casing **45** so that the cement swell packer **100** is set at a particular depth in the wellbore **20** (for example, a depth in which an additional or some cement seal is desired). When run into the wellbore **20**, the cementitious mesh packer **102** is exposed to wellbore fluids, such as a wellbore fluid circulated into the wellbore (including liquid as part of the primary cement **70** or a separate liquid circulated to the annulus **22**). In some aspects, wellbore fluids that are already present in the wellbore **20** cause swelling of the cementitious mesh packer **102**.

[0064] The cement swell packer **100** is installed at a final depth/location. In some aspects, the cementitious material **106** does not swell immediately on entry into the wellbore **20** due to, for example, built-in retarding agents within the material **106** (or generally, the cementitious mesh packer **102**). This can ensure that a diameter of the cementitious mesh packer **102** does not exceed, for example, the diameter, d , of the casing joint **50** for a particular time duration (for example, between 6 hours and 48 hour). Following this time period, the cementitious mesh packer **102** commences to swell by the hydration of the cementitious material **106**, which increases a diameter of the cementitious mesh packer **102**. Upon reaching a mechanical barrier, such as a wellbore wall, another casing string, or the primary cement **70**, the cementitious mesh packer **102** commences setting or building mechanical strength.

[0065] In some aspects, an acceptable strength of the cementitious mesh packer **102** is defined as the strength to ensure pressure holding capability across the cement swell packer **100**, which is obtained after a configurable range of 1 day to many weeks. For example, FIG. **5** is a graph **500** that shows relative strength development between a cement swell packer and a primary cement according to the present disclosure. Graph **500** includes x-axis **502** (of time, in hours) and y-axis **504** (of compressive strength in mega pascals). Curve **506** represents a compressive strength vs. time of a primary cement that is circulated into a wellbore (such as the primary cement **70**). Curve **508** represents a compressive strength vs. time of a cementitious material (such as material **106**) that is part of a casing swell packer (such as cement swell packer **100**).

[0066] As shown in graph **500**, five specific time instances and associated compressive strength values are identified for each curve **506** and **508**. The time instances are 0 hours, 6 hours (run in time of the cement swell packer), 24 hours (18 hours after run in of the cement swell packer and when a primary cement is circulated), 72 hours to 168 hours (curing time of both the cementitious material of the cement swell packer and the primary cement), and at 168 hours (7 days) the full pressure hold capability of both cement materials. The numbers on the curves **506** and **508** represent the compressive strength in mega pascals.

[0067] Graph **500**, therefore, shows an example deployment of a cement swell packer (such as cement swell packer **100**) in combination with a primary, pumped cement operation (in other words, primary cement **70**). Generally, the pumped cement is required to build acceptable (compressive) strength of 35 Mpa in 24 hours after deployment. With the cement swell packer starting to build strength 72 hours after deployment, then achieving a maximum of 17 Mpa, the example shows that the cement swell packer is unable to crush or otherwise damage pumped cement.

[0068] FIGS. **6A-6C** show schematic diagrams that illustrate a development over time of a cement swell packer in a wellbore according to the present disclosure. Generally, FIGS. **6A-6C** show schematic diagrams that illustrate a development over time (upon run in at 6 hours to 7 days after run in) of the cement swell packer **100** when there is no primary cement **70** circulated to the annulus **22** (in other words, behind the production casing **45** once installed in the wellbore). FIG. **6A** shows the condition of the location of the wellbore **20** in which the cement swell packer **100** is installed and swelling of the cementitious material **106** starts at 6 hours. FIG. **6B** shows the condition of the location of the wellbore **20** in which the cement swell packer **100** is installed and swelling or curing of the cementitious material **106** has been ongoing for 3 days. FIG. **6C** shows the condition of the location of the wellbore **20** in which the cement swell packer **100** is installed and swelling or curing of the cementitious material **106** has been ongoing for 7 days (for maximum achievable compressive strength). In this example, the cement swell packer **100** is configured to commence swelling after 6 hours. After 3 days, the cement swell packer **100** (i.e., cementitious material **106**) has swollen to fill the available annulus **22** and commences building mechanical strength. After 7 days, the cement swell packer **100** (i.e., cementitious material **106**) has cured or built sufficient strength (upon contact with the formation or an outer casing) to achieve the required pressure holding specification.

[0069] FIGS. **7A-7C** show additional schematic diagrams that illustrate a development over time of a cement swell packer in a wellbore according to the present disclosure. Generally, FIGS. **7A-7C** show schematic diagrams that illustrate a development over time (upon run in at 6 hours to 7 days after run in) of the cement swell packer **100** when there is a partial application of primary cement **70** circulated to the annulus **22** (in other words, behind the production casing **45** once installed in the wellbore). FIG. **7A** shows the condition of the location of the wellbore **20** in which the cement swell packer **100** is installed and swelling of the cementitious material **106** starts at 6 hours. FIG. **7B** shows the condition of the location of the wellbore **20** in which the cement swell packer **100** is installed and swelling or curing of the cementitious material **106** has been ongoing for 24 hours, with the partial primary cement **70** having been circulated into the annulus **22**. FIG. **7C** shows the

condition of the location of the wellbore **20** in which the cement swell packer **100** is installed and swelling or curing of the cementitious material **106** has been ongoing for 7 days (for maximum achievable compressive strength). In this example, the cement swell packer **100** is configured to commence swelling after 6 hours. After 3 days, the cement swell packer **100** (i.e., cementitious material **106**) has swollen to fill the available space of annulus **22** not taken up by the primary cement **70** and commences building mechanical strength. After 7 days, the cement swell packer **100** (i.e., cementitious material **106**) has cured or built sufficient strength to achieve the required pressure holding specification.

[0070] FIGS. **8A-8D** show additional schematic diagrams that illustrate a development over time of a cement swell packer in a wellbore according to the present disclosure. Generally, FIGS. **8A-8D** show schematic diagrams that illustrate a development over time (upon run in at 6 hours to 7 days after run in) of the cement swell packer **100** when there is a complete application of primary cement **70** circulated to the annulus **22** (in other words, behind the production casing **45** once installed in the wellbore). FIG. **8A** shows the condition of the location of the wellbore **20** in which the cement swell packer **100** is installed and swelling of the cementitious material **106** starts at 6 hours. FIG. **8B** shows the condition of the location of the wellbore **20** in which the cement swell packer **100** is installed and swelling or curing of the cementitious material **106** has been ongoing for 24 hours, with the primary cement **70** having been circulated into the annulus **22** and curing. FIG. **8C** shows the condition of the location of the wellbore **20** in which the cement swell packer **100** is installed and swelling or curing of the cementitious material **106** has been ongoing for 3 days; here, the cementitious material **106** is constrained when it contacts the primary cement **70** (which has a greater unconfined compressive strength than the cementitious material **106**) or another surface (such as another casing or wellbore wall). FIG. **8D** shows the condition of the location of the wellbore **20** in which the cement swell packer **100** is installed and swelling or curing of the cementitious material **106** has been ongoing for 7 days (for maximum achievable compressive strength). In this example, the cement swell packer **100** is configured to commence swelling after 6 hours. After 3 days, the cement swell packer **100** (i.e., cementitious material **106**) has swollen to fill the available space of annulus **22** not taken up by the primary cement **70** and commences building mechanical strength. After 7 days, the cement swell packer **100** (i.e., cementitious material **106**) has cured or built sufficient strength to achieve the required pressure holding specification.

[0071] While this specification contains many specific implementation details, these should not be construed as limitations on the scope of any inventions or of what may be claimed, but rather as descriptions of features specific to particular implementations of particular inventions. Certain features that are described in this specification in the context of separate implementations can also be implemented in combination in a single implementation. Conversely, various features that are described in the context of a single implementation can also be implemented in multiple implementations separately or in any suitable subcombination. Moreover, although features may be described above as acting in certain combinations and even initially claimed as such, one or more features from a claimed combination can in some cases be excised from the combination, and the claimed combination may be directed to a subcombination or variation of a subcombination.

[0072] A number of implementations have been described. Nevertheless, it will be understood that various modifications may be made without departing from the spirit and scope of the disclosure. For example, example operations, methods, or processes described herein may include more steps or fewer steps than those described. Further, the steps in such example operations, methods, or processes may be performed in different successions than that described or illustrated in the figures. Accordingly, other implementations are within the scope of the following claims.

Claims

- 1.** A cement swell packer, comprising: a base tubular that comprises at least one end connection; a cement swell assembly configured to mount on the base tubular and comprising: a portion of a dry cementitious material configured to form a hardenable or curable cement when exposed to a fluid in a wellbore; and a mesh configured to hold the portion of the dry cementitious material; and a liquid barrier formed over the dry cementitious material.
- 2.** The cement swell packer of claim 1, comprising at least one collar configured to retain the cement swell assembly on the base tubular.
- 3.** The cement swell packer of claim 2, wherein the at least one collar comprises a first collar configured to mount on the base tubular at a first end of the cement swell assembly and a second collar configured to mount on the base tubular at a second end of the cement swell assembly opposite the first end.
- 4.** The cement swell packer of claim 1, wherein the liquid barrier comprises a removable film or wrap.
- 5.** The cement swell packer of claim 1, wherein the liquid barrier comprises a sprayed barrier configured to degrade in the presence of the fluid.
- 6.** The cement swell packer of claim 1, comprising at least one centralizer configured to mount over the base tubular.
- 7.** The cement swell packer of claim 6, wherein the at least one centralizer comprises a first centralizer configured to mount on the base tubular at or near a first end of the base tubular and a second centralizer configured to mount on the base tubular at or near a second end of the base tubular opposite the first end.
- 8.** The cement swell packer of claim 1, wherein the base tubular comprises a casing joint of a production casing.
- 9.** The cement swell packer of claim 8, wherein the at least one end connection comprises: a male threaded connection at a first end of the casing joint; and a female threaded connection at a second end of the casing joint opposite the first end.
- 10.** The cement swell packer of claim 8, wherein an unconfined compressive strength of the hardenable or curable cement is less than an unconfined compressive strength of a primary cement pumped into the wellbore to adhere the production casing to a well surface.
- 11.** The cement swell packer of claim 10, wherein the well surface comprises an intermediate casing.
- 12.** A method, comprising: positioning a cement swell packer adjacent a wellbore entry, the cement swell packer comprising: a base tubular that comprises at least one end connection, a cement swell assembly mounted on the base tubular and comprising a portion of a dry cementitious material configured to form a hardenable or curable cement when exposed to a fluid in a wellbore; and a mesh configured to hold the portion of the dry cementitious material, and a liquid barrier formed over the dry cementitious material; removing the liquid barrier from the dry cementitious material; making up a connection between the at least one end connection and another base tubular within a string of base tubulars; and running the cement swell packer into the wellbore on the string of base tubulars.
- 13.** The method of claim 12, comprising: making up another connection between the base tubular and a further base tubular at an end of the base tubular opposite the another base tubular; and running the base tubular, the another base tubular, and the further base tubular into the wellbore on the string of base tubulars.
- 14.** The method of claim 12, wherein removing the liquid barrier from the dry cementitious material occurs subsequent to making up the connection between the at least one end connection and the another base tubular within the string of base tubulars.
- 15.** The method of claim 12, comprising: positioning the base tubular on the string of base tubulars at a particular location in the wellbore; and exposing the dry cementitious material to a liquid in the

wellbore to form a hardenable cementitious slurry.

16. The method of claim 15, comprising initiating a hardening or curing process of the hardenable cementitious slurry to form hardened cement over a time duration.

17. The method of claim 15, wherein exposing the dry cementitious material to the liquid in the wellbore comprises circulating the liquid from a terranean surface to the particular location in the wellbore.

18. The method of claim 17, wherein the liquid comprises water.

19. The method of claim 15, comprising: pumping a flow of a primary cement that comprises the liquid downhole through the string of base tubulars and uphole through an annulus between the string of base tubulars and a well surface; and exposing the dry cementitious material to the liquid in the wellbore to form the hardenable cementitious slurry.

20. The method of claim 19, wherein the well surface comprises an inner radial surface of an intermediate casing string, and the string of base tubulars comprises a production casing.

21. The method of claim 12, comprising assembling the cement swell packer by: mounting the cement swell assembly on the base tubular; installing at least one collar onto the base tubular to retain the cement swell assembly on the base tubular; and forming the liquid barrier onto the cement swell assembly.

22. The method of claim 21, wherein forming the liquid barrier onto the cement swell assembly comprises: covering the cement swell assembly with a film or wrapping to form the liquid barrier; or spraying the cement swell assembly to form the liquid barrier.

23. An apparatus, comprising: a plurality of casing joints, each casing joint comprising a tubular section, a male threaded connection, and a female threaded connection, the plurality of casing joints configured to form a connected production casing string in a wellbore; and a cement swell assembly mounted on at least one of the casing joints, the cement swell assembly comprising: a portion of a dry cementitious material configured to form a hardenable or curable cement when exposed to a fluid in the wellbore; and a mesh configured to hold the portion of the dry cementitious material.

24. The apparatus of claim 23, comprising: a first collar mounted on the at least one casing joint at a first end to retain the cement swell assembly on the casing joint; and a second collar mounted on the at least one casing joint at a second end of the cement swell assembly opposite the first end to retain the cement swell assembly on the casing joint.

25. The apparatus of claim 24, comprising: a first centralizer mounted on the casing joint at or near the first end; and a second centralizer mounted on the casing joint at or near the second end.

26. The apparatus of claim 23, comprising a liquid barrier that covers the portion of the dry cementitious material.

27. The apparatus of claim 23, wherein an unconfined compressive strength of the hardenable or curable cement is less than an unconfined compressive strength of a primary cement pumped into the wellbore to install the connected production casing string.
